

Rubber-Band Dynamics Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Valence Gap and Re-Alignment Analysis

PENPAL Project

2026-04-26

Contents

1 Overview	1
2 Load Data	2
3 Preprocessing	2
4 Primary: Per-Story Reversion Strength	2
5 Supplemental: Pooled Condition-Specific Slopes	3
5.1 Supplemental: Cross-Condition Interaction Model	4
6 Supplemental: General Gap Structure	5
7 Summary	6

1 Overview

This notebook compares **rubber-band dynamics** across conditions using the exchange-aligned data structure.

The main quantity is the **valence gap**:

- `valence_gap = slot_1_valence - slot_2_valence`
- `gap_change = current_gap - previous_gap`

A negative slope in `gap_change ~ previous_gap` indicates a rubber-band effect: when the slots diverge, the next aligned exchange tends to pull them back together.

As in the updated valence notebook:

- `author_1 / author_2` are treated as **slot 1 / slot 2**
- the analysis is restricted to **complete exchanges**
- the temporal window uses `analysis_turn / turn_index`, not the old raw-turn logic

Primary cross-condition object. For each story i , fit the OLS

$$\text{gap_change}_{i,t} = \alpha_i + \beta_i \cdot \text{valence_gap_lag1}_{i,t} + \varepsilon_{i,t}$$

and define the per-story reversion strength as $R_i = -\beta_i$, so that larger positive R_i means a stronger pull back together. R_i is already label-invariant (the gap enters squared/reversal-style in the relationship). We

test $R_i \sim \text{condition}$ with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon. The pooled LMM slopes are kept as supplemental.

2 Load Data

Table 1: Loaded valence data for rubber-band analysis

condition	n_stories	n_turns
Human-AI	100	851
Human-Human	36	324
AI-AI	38	342

3 Preprocessing

Analysis rows: 1343

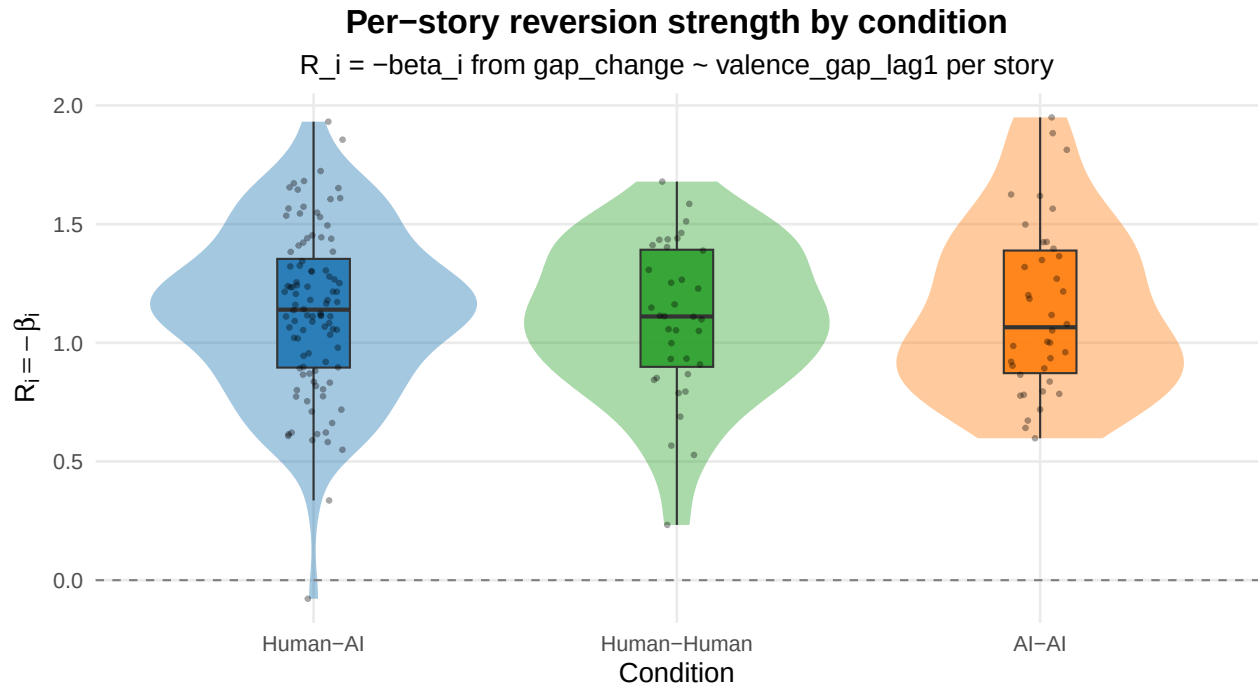
Stories: 174

4 Primary: Per-Story Reversion Strength

Table 2: Per-story reversion strength $R_i = -\beta$ by condition

condition	n	mean_R	median_R	sd_R
Human-AI	100	1.134	1.140	0.348
Human-Human	36	1.101	1.111	0.321
AI-AI	38	1.143	1.065	0.358

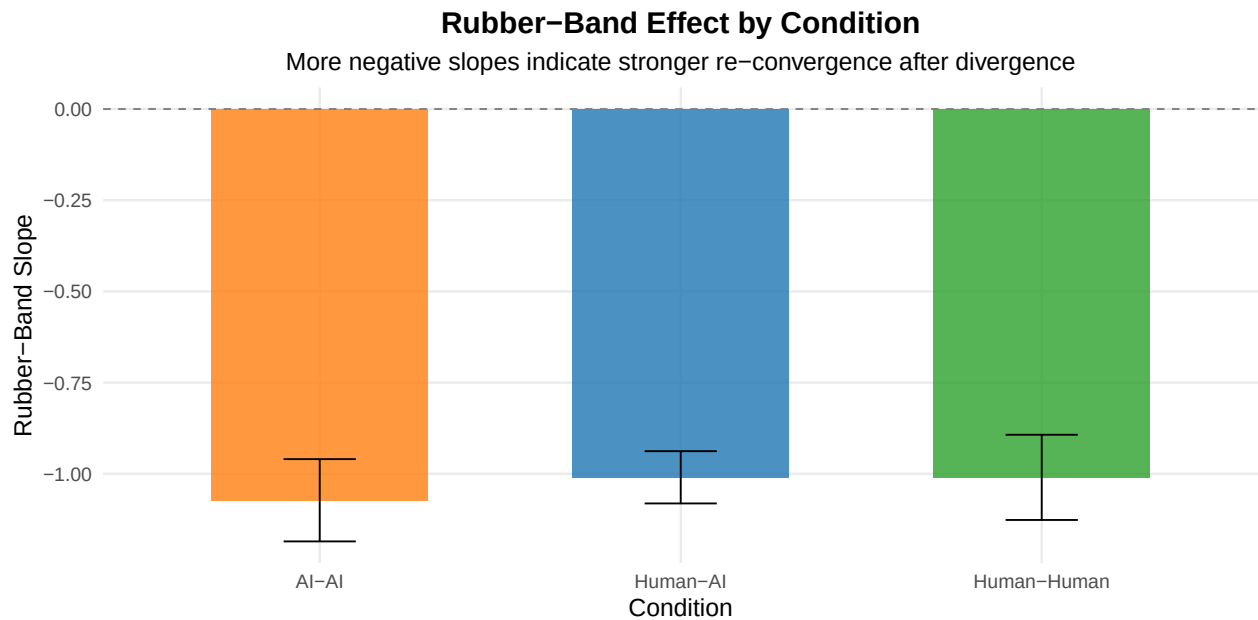
```
## One-way ANOVA: R ~ condition
##               Df Sum Sq Mean Sq F value Pr(>F)
## condition      2  0.038  0.01876   0.158  0.854
## Residuals    171 20.343  0.11896
##
## Kruskal-Wallis (robust): R ~ condition
##
## Kruskal-Wallis rank sum test
##
## data: R by condition
## Kruskal-Wallis chi-squared = 0.24702, df = 2, p-value = 0.8838
##
## Pairwise Wilcoxon (BH-adjusted):
##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data: per_story_rb$R and per_story_rb$condition
##
##           Human-AI Human-Human
## Human-Human 1         -
## AI-AI        1         1
##
## P value adjustment method: BH
```



5 Supplemental: Pooled Condition-Specific Slopes

Table 3: Rubber-band slopes by condition

condition	estimate	se	t	p	sig
Human-Human	-1.010	0.060	-16.945	0	***
Human-AI	-1.010	0.037	-27.619	0	***
AI-AI	-1.072	0.058	-18.640	0	***



5.1 Supplemental: Cross-Condition Interaction Model

```
## Cross-condition model summary:
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: gap_change ~ valence_gap_lag1 * condition + (1 | conversation_id)
## Data: df_analysis
##
## REML criterion at convergence: 1420.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.2433 -0.6217 -0.0158  0.6405  3.0612
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.006773 0.0823
## Residual                    0.159690 0.3996
## Number of obs: 1343, groups: conversation_id, 174
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)    -0.10520    0.01709 138.57099  -6.155
## valence_gap_lag1 -0.96509    0.03540 1312.74728 -27.262
## conditionHuman-Human    0.11845    0.03217 122.87815   3.682
## conditionAI-AI         0.10299    0.03156 123.17367   3.263
## valence_gap_lag1:conditionHuman-Human -0.07300    0.06528 1335.97636  -1.118
## valence_gap_lag1:conditionAI-AI    -0.12474    0.07898 1307.61321  -1.579
##
##              Pr(>|t|)
## (Intercept)    7.6e-09 ***
## valence_gap_lag1 < 2e-16 ***
## conditionHuman-Human 0.000345 ***
## conditionAI-AI      0.001426 **
## valence_gap_lag1:conditionHuman-Human 0.263662
## valence_gap_lag1:conditionAI-AI      0.114487
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) vln__1 cndH-H cAI-AI v__1:H
## vlnc_gp_lg1   0.199
## cndtnHmn-Hm  -0.531 -0.106
## condtnAI-AI  -0.542 -0.108  0.288
## vlnc__1:H-H  -0.108 -0.542  0.050  0.058
## vl__1:AI-AI  -0.089 -0.448  0.047  0.065  0.243
##
##
## Condition-specific slopes:
##      condition  valence_gap_lag1.trend      SE      df lower.CL upper.CL
## Human-AI        -0.965 0.0355 1320      -1.03   -0.896
## Human-Human     -1.038 0.0549 1328      -1.15   -0.930
## AI-AI           -1.090 0.0707 1292      -1.23   -0.951
##
## Degrees-of-freedom method: kenward-roger
```

```
## Confidence level used: 0.95
```

```
##
```

```
## Pairwise slope contrasts:
```

```
## contrast estimate SE df t.ratio p.value
## (Human-AI) - (Human-Human) 0.0730 0.0654 1336 1.117 0.7930
## (Human-AI) - (AI-AI) 0.1247 0.0791 1316 1.578 0.3446
## (Human-Human) - (AI-AI) 0.0517 0.0895 1309 0.578 1.0000
```

```
##
```

```
## Degrees-of-freedom method: kenward-roger
```

```
## P value adjustment: bonferroni method for 3 tests
```

6 Supplemental: General Gap Structure

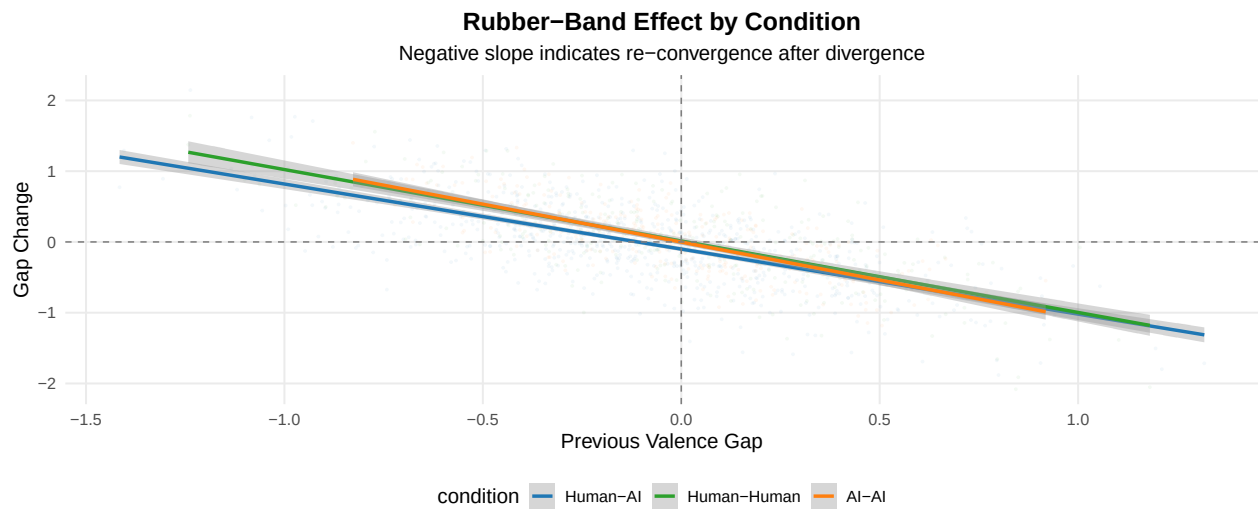
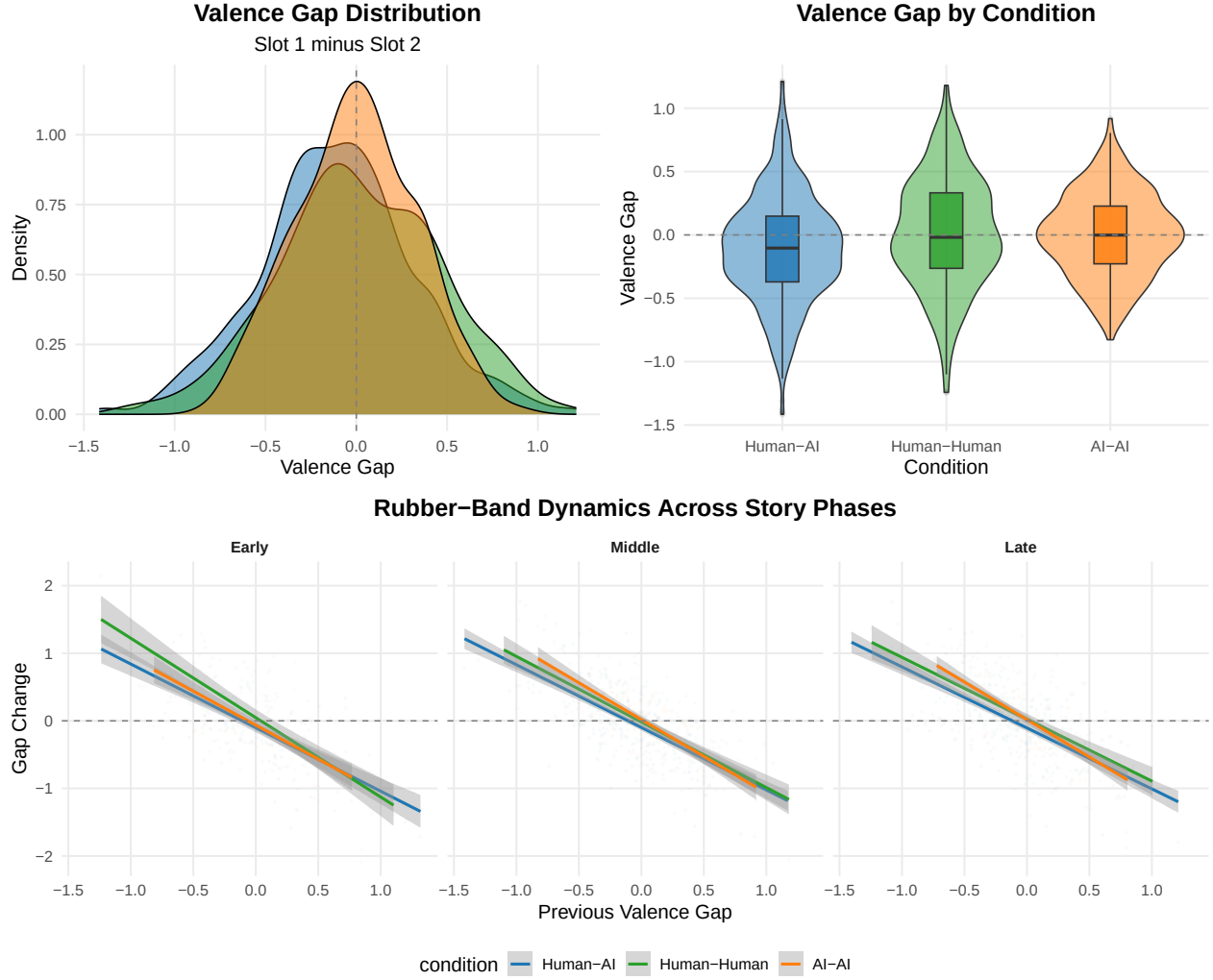


Table 4: Gap and gap-change descriptives by condition

condition	n_turns	mean_gap	sd_gap	mean_gap_change	sd_gap_change	rubber_band_r
Human-AI	751	-0.109	0.424	-0.013	0.574	-0.678
Human-Human	288	0.013	0.441	0.008	0.624	-0.708
AI-AI	304	-0.001	0.329	0.007	0.482	-0.731



7 Summary

Primary cross-condition result: per-story reversion strength $R_i = -\beta_i$ from `gap_change ~ valence_gap_lag1`, tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon.

Supplemental views retained:

- pooled within-condition LMM slopes (population-level rubber-band estimate per condition)
- cross-condition interaction LMM (`gap_change ~ valence_gap_lag1 * condition + (1|story)`)
- gap distributions and story-phase splits

Deferred to a follow-up pass:

- starter-mechanism analysis of the reversion object
- Human-AI decomposition of who pulls the gap back (requires per-turn speaker identity)

Analysis completed: 2026-04-26 22:10:42.094241